



2025

X23(B)

BOOKLET NO.

237026

Physics

Paper – II

Time Allowed : Three Hours

Maximum Marks : 250

Medium : English

Type of Paper : Conventional / Descriptive

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are EIGHT questions divided in two Sections, out of which FIVE are to be attempted.
2. Question No. 1 and 5 are compulsory. Out of the remaining, THREE are to be attempted choosing at least ONE question from each Section.
3. The number of marks carried by a question/sub-question is indicated against it.
4. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Keep in mind the word limit indicated in the question if any.
8. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
9. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
10. Neat sketches may be drawn, wherever required.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

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P.T.O.



SECTION – A

Q1. Solve the following.

(10×5=50)

Q1. (a) A free particle has the initial wave function

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$\Psi(x, 0) = Ae^{-a|x|}$ where A and a are positive real constants.

Then

i) Normalise $\Psi(x, 0)$

ii) Construct $\Psi(x, t)$ in the form of an integral.

[Given : Solution to the Schrodinger equation for the free particle is

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left(ikx - \frac{i\hbar}{2m} k^2 t\right) \tilde{\Psi}_0(K) dK$$

Q1. (b) If the Hamiltonian $H(\lambda)$ of a quantum system be a function of a continuous parameter ' λ '. If $E_n(\lambda)$ be the n^{th} eigenvalue and $|\Psi_n(\lambda)\rangle$ be its corresponding normalised eigen state such that $H(\lambda)|\Psi_n(\lambda)\rangle = E_n(\lambda)|\Psi_n(\lambda)\rangle$

Show that derivative of eigenvalue E_n with respect to the parameter ' λ '

$$\text{is given by } \frac{\partial E_n(\lambda)}{\partial \lambda} = \left\langle \Psi_n(\lambda) \left| \frac{\partial H(\lambda)}{\partial \lambda} \right| \Psi_n(\lambda) \right\rangle.$$

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Q1. (c) Evaluate the following commutators

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i) $[\bar{L}, \bar{S}, \bar{L}]$

ii) $[\bar{L}, \bar{S}, \bar{S}]$

iii) $[\bar{L}, \bar{S}, \bar{J}]$

iv) $[\bar{L}, \bar{S}, L^2]$

v) $[\bar{L}, \bar{S}, S^2]$

vi) $[\bar{L}, \bar{S}, J^2]$

Comment on your result.



- Q1. (d) What do you mean by “term symbol” ? What is meant by L–S coupling ? 10
- i) Derive the ground state term for free ion with d^9 .
 - ii) Find L, S and J of 3F and 1S .
 - iii) Determine J energy level (J states) for the term symbols for P^2 configuration.
- Q1. (e) Pauli’s spin matrices are represented by $\hat{\sigma}_x$, $\hat{\sigma}_y$ and $\hat{\sigma}_z$. Show that each has eigenvalues +1 and –1. 10
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- Q2. (a) The stationary states for an infinite square well potential given by following equation $V(x) = \begin{cases} 0 & \text{if } 0 \leq x \leq a \\ \infty & \text{otherwise} \end{cases}$ are found to be
- $$\psi_n(x, t) = \sqrt{\frac{2}{a}} \exp\left(-i \frac{\hbar \pi^2 n^2}{2ma^2} t\right) \sin \frac{n\pi x}{a}.$$
- The expectation value $\langle x \rangle$ for the $\psi_n(x, t)$ is found to be $a/2$ and $\langle p \rangle$ is found to be 0.
- Calculate following for $\psi_n(x, t)$
- $\langle x^2 \rangle$, $\langle p^2 \rangle$, σ_x , σ_p for the n^{th} stationary state. Check that the uncertainty principle is satisfied. Also find out which state comes closest to the uncertainty limit. 20
- Q2. (b) Apply ‘Zeeman effect’ to the Hydrogen atom. 20
- i) Calculate Hydrogen atom’s internal magnetic field due to orbital motion of the electron. Consider the Hamiltonian for a charged particle in a uniform magnetic field $H = \frac{p^2}{2M} - \frac{q}{2M} \bar{B}_0 \cdot \bar{L} + \frac{q^2}{8M} (\bar{r} \cdot \bar{r})(\bar{B}_0 \cdot \bar{B}_0) - (\bar{r} \cdot \bar{B}_0)(\bar{r} \cdot \bar{B}_0) + q\phi$.
 - ii) Explain what do you mean by weak field Zeeman effect. Derive an equation for Lande’s ‘g’ factor. What is Zeeman contribution to the energy ?
 - iii) Comment on your result.
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- Q2. (c)** Explain the three-dimensional free electron model in terms of Pauli's exclusion principle. Consider a solid metal cube of edge length 2.0 cm. **10**
- What is the lowest energy level for an electron with the metal ?
 - What is the spacing between this energy level and next energy level ?

$$\left(\begin{array}{l} \text{mass of electron} = 9.11 \times 10^{-31} \text{ kg} \\ \hbar = 1.05 \times 10^{-34} \text{ J.s} \end{array} \right)$$

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- Q3. (a)** Calculate the transmission, probability 'T' for a particle of mass 'm' incident from left with energy E in the limit $T \ll 1$. How wide must the barrier so that the transmission probability is 10^{-3} when an electron with energy $E = 1 \text{ eV}$ is incident upon the rectangular barrier of potential energy $V_0 = 2 \text{ eV}$?
($\hbar = 6.582 \times 10^{-16} \text{ eV.s}$, mass of electron = $9.1 \times 10^{-31} \text{ kg}$) **20**

- Q3. (b)**
- Explain in detail process of Raman scattering. Use quantum mechanical framework. **20**
 - What is Raman spectroscopy and what is Raman spectrum ? What information can be extracted from Raman spectrum ? Draw Raman spectrum of a diamond using arbitrary units on both axes. Justify your answer.
 - What are Raman images ? How do they differ from Raman spectra ?

- Q3. (c)** State the Franck-Condon principle and give its wave-mechanical interpretation. How does it help in understanding the intensity distribution in the vibrational structure of the electronic transitions of diatomic molecule ? **10**
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- Q4. (a) A particle of mass 'm' and kinetic energy $E > 0$ approaches an abrupt potential drop V_0 , what is the probability that it will reflect back if $E = V_0/3$? 20
- Q4. (b) Explain in detail the phenomenon of fluorescence and phosphorescence using Jablonski diagram. Discuss in detail the various deactivation processes. Compare the photon absorption and emission rates and time scales. What do you mean by quantum efficiency? Compare between quantum efficiency of fluorescence and phosphorescence. 20
- Q4. (c) What is the origin of fine structures of Hydrogen atom? What are the three perturbations applied to the Schrodinger Hamiltonian for proper analysis of fine structures? Find the lowest order relativistic correction to the energy levels of one-dimensional harmonic oscillator. 10
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SECTION – B

- Q5. Solve the following. (10×5=50)
- Q5. (a) What are the salient features of nuclear forces? Briefly explain the Meson theory of Nuclear forces (Yukawa theory) and the role of π -meson? Explain the difference between binding energy and nuclear force. 10
- Q5. (b) What do you mean by nuclear wavefunction? Explain the physical significance of parity and angular momentum as basic nuclear properties. Give the rules for determining the parity of a nuclear state based on Shell model. 10
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- Q5. (c)** What do you mean by FCC, BCC and simple cubic structure ? The atomic weight, density and atomic radius for three hypothetical alloys are listed in the following table. For each, determine whether its crystal structure is FCC, BCC or simple cubic and then justify your determination. **10**

Alloy	Atomic Weight (g/mol)	Density (g/cm ³)	Atomic Radius (nm)
A	77.4	8.22	0.125
B	107.6	13.42	0.133
C	127.3	9.23	0.142

- Q5. (d)** Discuss Debye's contribution to the specific heat theory. How it improved Einstein's oscillator treatment of specific heat ? **10**
- Q5. (e)** What is the significance of Mass Parabola in nuclear physics ? Use the Semi Empirical Mass Formula (SEMF) to explain the concept of 'β -stability' and how it determines the nature of β-decay for a given isobaric chain ? **10**

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- Q6. (a)** Discuss the nature of Deuteron bound state. Explain how the experimental observation of its magnetic dipole moment and the existence of non-zero electric quadrupole moment provide evidence for the presence of non-central (tensor) forces in the nuclear interaction. **20**

- Q6. (b)** Explain formation of bands in solids. With the help of suitable diagrams define valence band, conduction band and forbidden gap. How do you classify solids on the basis of band theory ? Compare the number of conduction electrons in Germanium at temperature 45°C and 27°C.

[Given $E_g = 0.72$ eV, $m_e^* = m_e = 9.1 \times 10^{-31}$ kg, $k = 8.62 \times 10^{-5}$ eV/K] **20**

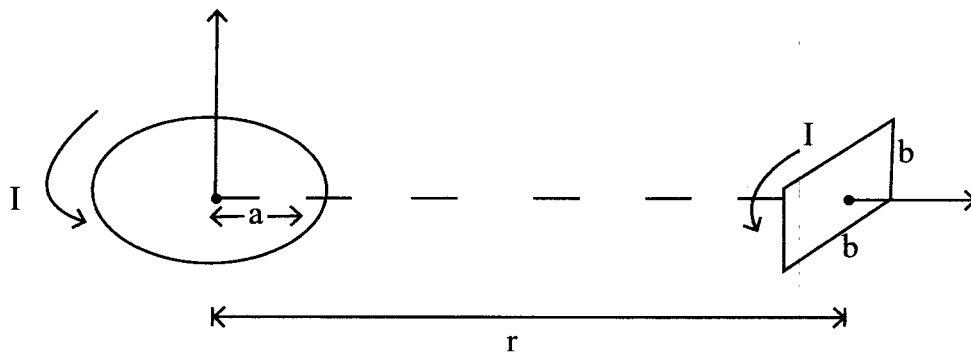
- Q6. (c)** Briefly describe the mechanism of nuclear fusion and explain its role as the primary source of energy production in stars. **10**
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Q7. (a) Classify the elementary particles based on their interactions and spin. Briefly discuss the field Quanta of electroweak and strong interactions, highlighting the concept of conservation laws in particle physics. **20**

Q7. (b) Calculate the torque exerted on the square loop (each side dimension = b) due to circular current carrying loop of radius a , which is at a distance r ($r \gg a, b$).

What will be its equilibrium orientation ? **20**



Q7. (c) i) What do you mean by JFET ? What are its salient features ? How it differs from 'bipolar transistor' ? Explain how JFET can be used as an amplifier ? **10**

ii) The transconductance of JFET used as a voltage amplifier is $3000 \mu\text{mho}$ and drain resistance is $10 \text{ k}\Omega$. Calculate the voltage gain of the amplifier.

iii) If a $4.7 \text{ k}\Omega$ load resistor is a.c. coupled to the output of the JFET amplifier, what is the resulting r.m.s. output voltage ?

[Given $g_m = 779 \times 10^{-6} \text{ S}$, $R_D = 3.3 \text{ k}\Omega$, $V_{in} = 100 \text{ mV}$.]



- Q8. (a) Discuss the shell model of the nucleus. Describe its success in explaining nuclear properties. What are the major limitations of this model ? 20
- Q8. (b) i) What is micro processor ? What are the core components of the microprocessor ? Describe briefly function of each component with the help of suitable diagram. 20
- ii) Differentiate between microprocessors 8085 and 8086.
- iii) Write an assembly language program in 8085 microprocessor to find maximum and minimum of 10 numbers considering following assumption
Assumption – List of numbers from 2050 H to 2059 H
Output at 2060 H and 2061H.
- Q8. (c) Explain the difference between X-Ray Diffraction (XRD) and Transmission Electron Microscopy (TEM) technique for material characterisation. Further, explain how they are complementary to each other. 10

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